Leftover Bytes

(MVP, Prototype Strategy & Wireframes)

Team Members (Product Managers)

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Minimum Viable Product (MVP)

Student Login (Kerberos)

Leverages BU Kerberos credentials to auto-authenticate students and organizers securely.

Dietary Preference Setup

Uses NLP-backed classification of uploaded food items (from organizer photos or checkboxes) to match against user preferences: vegan, gluten-free, halal, nut-free, etc.

Smart Push Notifications

Combines GPS and Wifi triangulation with Firebase push to notify only nearby students when food becomes available, reducing irrelevant alerts.

Event Activation System

Organizers upload pictures with time and location tagging; students can preview food before heading over.

Gamified Feedback & Status Updates

Students can “mark as picked,” upload new pics, or leave freshness notes to earn eco-points—exchangeable for rewards like reusable utensils or sustainability badges.

Dashboard for Organizers

Post-event analytics show how many portions were picked up, waste prevented (in lbs), and impact score (visible to Zero Waste Office).

Minimum Viable Product (MVP)

✅ Included in MVP	❌ Not Included in MVP
Student Login (Kerberos)	Social media sharing
Dietary Preference Setup	Real-time pickup tracking
Initial Food Notifications	Inventory tracking
Event Activation System	Multi-campus rollout
Photo uploads & Point system	Admin moderation tools
Leaderboard + Certifications	QR scan-based verification

User Stories & Functional Requirements

Primary User: BU Students

"As a student juggling classes and a tight budget, I want to get alerts for free food nearby that fits my diet so I can stay nourished without stress."

"As someone who cares about sustainability, I want to feel like I'm part of a solution, not just a beneficiary."

As a student, I want to...

- Get real-time alerts when leftover food is nearby
- Receive notifications based on dietary preferences
- See food pictures and location before heading out
- Earn points by uploading status updates
- Be recognized on a leaderboard for participation

Secondary User: Event Organizers

"As someone wrapping up a long event, I need a simple way to share leftover food and know it's being picked up—without sticking around for 30 minutes."

"I want to feel like I made a difference with minimal extra work."

As an event organizer, I want to...

- Quickly post leftover food via my phone
- Avoid staying back to manually distribute food
- Ensure the food reaches students efficiently

Low-Fidelity Prototype

Core Experience Flow

Sign-In
Log in with BU Single Sign-On.

↓

Set Preferences
Auto-fetch dietary preferences or set up manually.

↓

Home Feed
Receive food alerts ranked by distance.

↓

Detail View
Review food details, estimated travel time, and images.

↓

Pick Up & Update
Collect food, snap leftover pic. Update status and win rewards.

↓

Rewards
Earn points, view leaderboard, and get certificates/goodies.

Organizers Flow

Sign-In
Sign in via Work Email

↓

Create Post
Upload image + select food types from icons

↓

Send Notification
Auto Notify Nearby Students

↓

Event Dashboard
Receive food status

Wireframe mockups will show both mobile and desktop sketches to simulate how the platform is equally accessible across devices.

AI Automation and Augmentation

AI Automation & Augmentation Strategy:

We're not just building an app—we're building a context-aware food sharing network with AI as the invisible co-pilot.

AI-Driven Features:

Smart Timing Engine
Learns when notifications are most effective (e.g., lunch rush vs. late-night events), optimizing push alert schedules dynamically.

Auto-Categorization of Food Types:

Uses AI vision APIs (like Google Cloud Vision or AWS Rekognition) to scan uploaded food pics and recommend dietary tags automatically.

Location-Aware Sorting:

Hybrid geolocation logic ranks food alerts based on proximity, crowdedness level, and accessibility (e.g., elevators available for disabled students).

Predictive Waste Modeling:

Analyzes past event data, time, and type (e.g., student org vs. external speaker panel) to forecast surplus and prompt early action.

Feedback Collection Framework

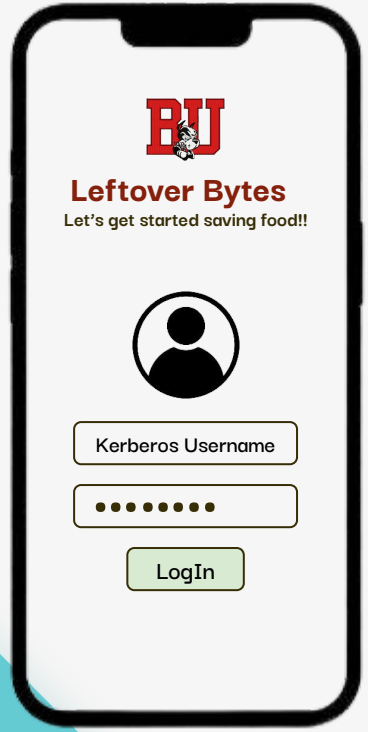
	01	02	03	04
	Pre-MVP Phase (now)	Pilot Test Phase (CDS events)	Beta Rollout	Post-MVP Phase
Goal	Validate key assumptions	Usability testing & engagement	Feature-specific feedback	Scalability
Method	User interviews (students, organizers, Zero Waste staff)	QR code surveys, opt-in analytics, intercept interviews	A/B tests for gamification, leaderboard impact	Weekly engagement metrics, drop-off heatmaps, NPS tracking

Feedback Collection Framework

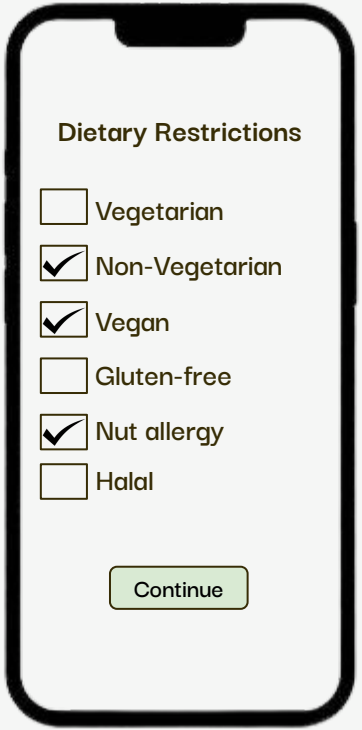
Stakeholder	Method	Insight
Students with dietary needs	🧠 Think-Aloud Testing 📊 Survey Forms 🃏 Card Sorting	Usability pain points, emotional reactions, feature priorities
Event Organizers (BU staff, clubs)	🧪 Organizer Pilot Simulation 📊 Survey Forms	Ease of post-event uploading, dashboard relevance
Sustainability Office	🧠 Think-Aloud Testing 📊 Live Polls	Environmental relevance, priority feedback
BU Dining & Facilities	🧠 Think-Aloud + Dashboard Simulation 📊 Q&A Feedback	Operational fit, practical use of filters/stats

Wireframe: BU Students

Login Page

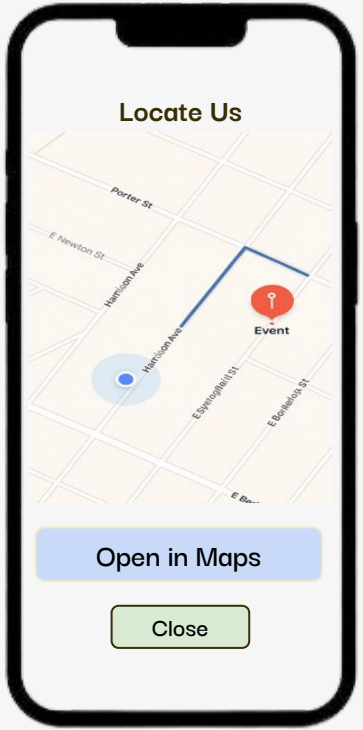


Preferences



Event Calendar

Location



Wireframe: BU Students

Event Dashboard

Event 1: Earth Day

FIRST COME, FIRST SERVE

Pizza

Vegan: 10

NonVeg: 5

Pickup At:
CDS 201

Upload image

Upload Image

Upload Image

(2 pts per Photos)

Upload

Earn Rewards

(Get by uploading Pics)

Check Points

Leaderboard

Leaderboard

Name	Points
Alen	20
Ram	11
William	5

Get Rewards

Rewards Page

Certifications & Goodies

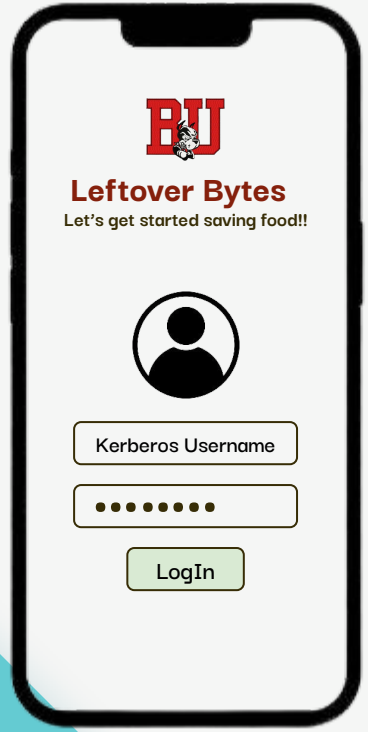
(or)

Claim

Sign out

Wireframe: Event Organizers

Login Page



Upload Image

Preferences

Events Conformation

Thank You!